**12** In an experiment 500 fruit flies were released into a controlled environment. After 10 days there were 650 fruit flies present.

Munirah believes that $N$, the number of fruit flies present at time $t$ days after the fruit flies are released, will increase at the rate of 4.4% per day. She proposes that the situation is modelled by the formula $N = Ak^t$.

**(i)** Write down the values of $A$ and $k$. **[2]**

**(ii)** Determine whether the model is consistent with the value of $N$ at $t = 10$. **[2]**

**(iii)** What does the model suggest about the number of fruit flies in the long run? **[1]**

Subsequently it is found that for large values of $t$ the number of fruit flies in the controlled environment oscillates about 750. It is also found that as $t$ increases the oscillations decrease in magnitude.

Munirah proposes a second model in the light of this new information.

$$N = 750 - 250 \times e^{-0.092t}.$$

**(iv)** Identify three ways in which this second model is consistent with the known data. **[3]**

**(v)** (*A*) Identify one feature which is not accounted for by the second model. **[1]**

(*B*) Give an example of a mathematical function which needs to be incorporated in the model to account for this feature. **[1]**